

Load balanced calls and RestTemplate implementation

Spring Cloud uses Netflix Ribbon, a client-side load balancer that plays a critical role and can handle both HTTP and TCP protocols. Ribbon is cloud-enabled and provides built-in failure resiliency. Ribbon also allows you to use multiple and pluggable load balancing rules. It integrates clients with load balancers.

Ribbon is integrated with the Eureka Server for client-side load balancing and with the Zuul server for server-side load balancing in Spring Cloud by default. This integration provides the following features:

- There is no need to hard code remote server URLs for discovery when Eureka Server is used. This is a prominent advantage, although you can still use the configured server list (`listOfServers`) in `application.yml` if required.
- The server list gets populated from Eureka Server. Eureka Server overrides `ribbonServerList` with `DiscoveryEnabledNIWSServerList`.
- The request to find out whether the server is up is delegated to Eureka. The `DiscoveryEnabledNIWSServerList` interface is used in place of Ribbon's IPing.

There are different clients available in Spring Cloud that use Ribbon, such as `RestTemplate` or `FeignClient`. These clients allow microservices to communicate with each other. Clients use instance IDs in place of host-names and ports for making an HTTP call to service instances when Eureka Server is used. The client passes the service ID to Ribbon; Ribbon then uses the load balancer to pick the instance from the Eureka Server.

If there are multiple instances of services available in Eureka, Ribbon picks only one for the request, based on load balancing algorithms. There

are two instances of `restaurant-service` registered on Eureka Server, as shown in the following screenshot:

Instances currently registered with Eureka			
Application	AMIs	Availability Zones	Status
RESTAURANT-SERVICE	n/a (2)	(2)	UP (2) - SOUSHARM-IN:restaurant-service:5b034f31fd44c9ff6dd5c5fb1d4c83d7} , SOUSHARM-IN:restaurant-service:707b060d8d02e3516f3fde3c86c858d1}
ZUUL-SERVER	n/a (1)	(1)	UP (1) - SOUSHARM-IN:zuul-server:9094e5aae179efe903061d827e21e167}

Now, we'll explore how you can make the REST calls from one service to another using `RestTemplate`.